

Exact Audits of Worked Systems for the Obstruction Calculus: Counts, Witnesses, and Monitoring Design Rules

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Abstract

Viability certificates are useful in applications only insofar as their behaviour on concrete systems is understood precisely. This paper develops one exact audit framework — integer and rational arithmetic throughout, every count regenerated by script — and applies it to a family of worked systems that instantiate the mechanisms of the obstruction calculus : a shared finite two-coordinate system with observation structures (aggregation, policy-class restrictions, decentralized observation, regime uncertainty), a review-timing grid for a hidden-regime resource model, and a continuous two-floor production benchmark. The audits yield the following. The review-timing identity on the 93-cell grid is non-strict: viability at the deadline holds if and only if $T_{\text{obs}} \leq \tau^*(z_0) = z_0 - 1$, boundary included; the strict form against the real exit bound fails exactly on the boundary cells (against the floored count σ^* , on one cell per row from $z_0 = 2$). On the four-state monitoring target, 7 of the 15 partitions are adequate with two incomparable coarsest adequate two-cell partitions (minimal in the refinement order — hence no single coarsest adequate design exists), and the two maximal common-action sets overlap. Four policy classes have kernel sizes 24, 26, 25, and 28 of 36 belief pairs — the classes form a two-by-two product of aggregation level and protocol — with incomparable middle cells. Decentralized unanimity over a 256-law space (64 behaviorally distinct) sustains 12 of 36 pairs; a whole-law authority restriction collapses the kernel to zero, and either agency’s low vote forced against the target kills it. Under frozen regimes no pair is lost (the corner pins its state); mode-blind operation loses exactly the pairs containing the corner state (1, 1). The biased certainty-equivalence law has drift at least 21/100 on [1, 2]; the bias-free law has drift identically zero. The benchmark’s critical aggregate is $Y^* = 27/5$ (feasible, margin 0); at the audited obstructed aggregate $Y = 6$ a rational dual certificate prices the shortfall at margin 3/50. Static duality holds at value $-1/10$ on the two-floor instance with an ℓ_1 -normalized Farkas pair, and a Helly-tight triple exhibits the sparse $m+1$ -constraint witness. The complete result tables are carried here: the master table of all 36 belief pairs against the five kernels, the sixteen sustaining law pairs of the decentralized protocol (an exact 4×4 product), the 93-cell timing grid, the fifteen-

partition census with each fibre’s common safe-action set, and the benchmark and drift tables, with a five-figure set regenerated from the same scripts. Design rules follow: review deadlines are boundary-inclusive; merged fibres call for splits, not threshold tuning; aggregated indices support certainly-safe reporting only; biased observation maps call for structural correction.

Keywords: viability kernels; obstruction certificates; exact arithmetic; monitoring design; decentralized observation; certainty equivalence; max-plus algebra

1. Introduction

The obstruction calculus states sound sufficient conditions for nonviability — obstruction certificates — mechanism by mechanism: common-action failure (Sections 5–9), review-timing failure (Section 8), finite-time exit under regime alternation (Section 10), and failure induced by restricting the policy class or by a biased observation map (Sections 6, 13) (Abaee, 2026, An obstruction calculus). Each mechanism is accompanied in application by quantitative questions. How many belief states does an observation structure destroy, and which ones? At which review deadlines does the timing certificate fire, and does it fire at the boundary? What exactly does a vote protocol cost, and which law in the protocol’s space sustains a given target? This paper answers such questions exactly for a family of worked systems, under one audit frame and one notation. The sharp thesis, made explicit by the master theorem of Section 4: obstructions arise from five mechanisms — common-action failure, memory/register coupling, finite-time boundary exit, decentralized information loss, and joint convex infeasibility — and each admits a certificate with an explicit semantic scope, a monotonicity status, and a robustness margin.

Three properties define the frame. First, all computations are carried out in integer and rational arithmetic: no solver, no tolerance, no simulation. Second, every discrete census in the family is finite — the continuous objects reduce to exact rational calculations on rationally specified domains — and every count is regenerated by a deterministic standard-library script, so any two executions agree. Third, each audited quantity is stated together with a witness or a certificate — an action sequence, a partition, a

law pair, a dual measure — so the counts carry their own evidence. This paper carries the audit’s complete result tables and its figure set: the per-pair membership table behind the kernel counts, the sustaining law pairs behind the decentralized count, the full timing grid, the full partition census, and the benchmark and drift tables, every entry of which is regenerated from the same scripts that verify the text.

The systems are drawn from the applied model of Abaee (2026, Robust viability of the 2J3KL limit reference point), in which a two-coordinate resource stock is managed by acting one of two instruments under incomplete observation. The family comprises: the shared audit system with its observation structures (Sections 5–7 and 9–10); the hidden-regime review grid (Section 8); a continuous two-floor production benchmark (Section 11); and a multiple-floor line extending the Farkas certificate to jointly binding constraints (Section 14). The shared audit system is the running instance on which the paper’s counts are developed; each further member of the family is introduced against it. The contributions are: the master monotonicity theorem — one audited-tuple object whose kernel comparisons unify the policy, observation, and memory enlargements (Section 4); the complete exact audit of the worked family — the master table, the decentralized count, the full timing grid, the partition census, and the benchmark and drift tables, every entry regenerated by the verification scripts (Sections 5–10); and the design rules the audits license — boundary-inclusive review deadlines, merged-fibre caution, the monitoring-adequacy criterion, and the five-mechanism obstruction taxonomy with its certificate semantics (Sections 8–9).

2. Related work

Viability, and why worked systems. That the governing question is which states admit controls keeping every constraint satisfied over time, rather than which trajectory optimises an objective, is the founding move of viability theory (Aubin, 1991). The obstruction calculus audited here is the observation-constrained development of that programme, and the systems below instantiate its mechanisms. Certificates are useful in application only insofar as their behaviour on concrete systems is understood exactly, which is the purpose of this paper: every count is regenerated by script in integer and rational arithmetic.

Worked benchmarks in the verification community. The practice of maintaining a shared, precisely specified family of benchmark problems on which competing methods are compared is well established in the formal-verification community, where an annual friendly competition applies a changing set of tools to a fixed set of benchmark problems in continuous and hybrid systems, with results reported per benchmark rather than as a single ranking (Althoff et al., 2020; Geretti et al., 2020). This paper’s worked-systems family plays the analogous role for the obstruction calculus: a fixed set of instances whose exact answers are known and recomputable, on which the behaviour of the machinery can be read off.

The difference is what is being certified. The ARCH benchmarks are reachability problems for continuous and hybrid dynamics, solved by reachability tools; the in-

stances here are *information-constrained* viability problems under partial observation, and the quantities tabulated are counts of survivable belief pairs, kernel sizes under policy-class restrictions, and the sizes of minimal witness sets. No reachability tool computes these.

Computing viability kernels. The viability kernel is the object several of the audits tabulate, and it has a computational literature that should be named even though its setting differs. Saint-Pierre’s algorithm approximates the kernel by a sequence of backward-reaching sets with a provable convergence property (Saint-Pierre, 1994); support-vector representations approximate the kernel and the associated resilience values from simulated trajectories, with the classifier standing in for the indicator function of the kernel (Deffuant, Chapel and Martin, 2007). Both address *fully observed* dynamics in continuous or large discrete state spaces, where exact computation is infeasible and the honest goal is a guaranteed approximation.

The audits here sit at the other end of that trade. The state spaces are small enough that the kernel is computed *exactly*, in integer and rational arithmetic, so there is no approximation error to bound; and the objects are not Saint-Pierre’s kernels but their observation-constrained counterparts — kernels over *belief pairs* under declared observation structures and policy-class restrictions, which is why the tabulated quantities are counts of survivable pairs rather than subsets of a state space. Neither algorithm above computes those, since both presuppose that the controller sees the state. What transfers is the *question* — which states admit a control keeping every constraint satisfied — and the observation refinement is monotone when the policy and register semantics are held fixed; comparisons changing several of those axes at once need not be ordered.

The classical instruments the audits exercise. Three classical results recur and are named where they are used: the alternative for systems of linear inequalities underlying the duality certificate (Farkas, 1902); Helly’s theorem on intersecting convex sets, which governs the tightness of the sparse witness exhibited below (Helly, 1923); and max-plus algebraic methods for the discrete-event structure of the review-timing grid (Baccelli, Cohen, Olsder and Quadrat, 1992). None of these is contributed here; what is contributed is the exact audit of what they do on each instance.

Exact and interval computation. The audits run in integer and rational arithmetic throughout. This is a statement about the verification pipeline and not a claim of superiority over validated or interval-arithmetic approaches: the choice buys exactness at the cost of scope, and the instances below are deliberately small enough to pay that price.

3. Systems, conventions, and methods

The shared audit system. States are pairs $x = (z_1, z_2) \in \{0, 1, 2, 3\}^2$; the safe set is $\mathcal{V} = \{z_1 \geq 1, z_2 \geq 1\}$ with the two coordinates as floors. Instruments are $u \in \{0, 1, 2\}$: $u = j \in \{1, 2\}$ acts coordinate j , which regenerates one

unit up to the cap 3, while the other coordinate loses one unit; $u = 0$ executes no instrument and both coordinates lose one. The transition is defined index-wise: the acted coordinate regenerates, $F_j(z)_j = \min\{3, z_j + 1\}$, and the other declines, $F_j(z)_{3-j} = \max\{0, z_{3-j} - 1\}$, for $j \in \{1, 2\}$; $F_0(z)_i = \max\{0, z_i - 1\}$ for both coordinates i — coordinates clip at zero, the cap is saturating — and safety is evaluated after each transition. A coordinate-swapped misreading (the regenerated coordinate written first for both j) reproduces no audited count and is excluded by the verification script. Instrument 0 is never uniquely viable on the audited family: admitting $u = 0$ as a third action leaves all four kernel counts unchanged (24, 26, 25, 28; verified by the verification script, which also checks the structural premises — $F_0 \leq F_1$ and $F_0 \leq F_2$ pointwise on the grid and the safe set an up-set, the one-step reason no fibre needs 0). The corner (1, 1) is the single state from which no instrument maintains both floors, so its singleton is nonviable; the nine safe states generate $C(9, 2) = 36$ two-element belief pairs, the audited population (28 of the pairs full-information-viable; column F of Table 1), at review horizon $H = 12$ under the audited two-horizon verdict (the winning recursion at H and at $H - 2$, which agree setwise on the audited universe and on the whole 2^9 -belief universe (stabilized from horizon 2 there, and from horizon 1 for the two non-alternation classes) — $W_{12} = W_{10}$ for each policy class, verified by the verification script — so the verdict certifies a stabilized recursion; Abaee, 2026, An obstruction calculus, Theorem 1).

Observation structures. An observation map $\text{info} : x \mapsto c$ partitions a belief into fibres; a policy acts per fibre on the read cell. The audited policy classes combine two aggregation levels (full information; aggregation $c(x) = z_1 + z_2$) with two protocols (free per-step choice; the no-repeat protocol that forbids repeating the previous action. The protocol’s register is formalized as follows: one previous-action register serves each review, shared by all fibres of the belief and not one per branch; the policy fixes one action per fibre, the register then holds some fibre’s executed action, and the audited rule is pessimistic — the belief survives only if every admissible thread (each fibre’s action entering the register) leaves a winning continuation. Register values are the two instruments, the initial register is unconstrained, and the survival condition quantifies over every admissible thread, so the semantics is thread-independent by construction (Proposition 5 computes what single-thread rules and per-branch registers give instead).

Exactness. Counts are computed by backward recursion over fibre partitions with exact state arithmetic; witnesses are extracted as step-one per-fibre action vectors along a winning branch; the certificate itself is the recursion invariant — winning-set membership at every horizon level, quantified universally over adversarial successor states and register threads — of which the archived step-one vector is the first level. The census of Section 9 enumerates all 15 partitions of a four-state target; partitions are ordered by refinement — P is coarser than P' when every fibre of P' lies in a fibre of P — and “coarsest adequate” is relative to this order. The continuous objects of Section 11 are rationals.

Conventions. On the hold class of the hidden-regime model, the worst-case exit time is $\tau^*(z_0) = z_0 - 1$ and the

discrete survival supremum (the largest admissible integer review count) is $\sigma^*(z_0) = \lfloor z_0 - 1 \rfloor = \lfloor \tau^*(z_0) \rfloor$, with level sets $\{z_0 \geq 1 + k\}$; for an integer deadline the inclusive criterion $T_{\text{obs}} \leq \tau^*(z_0)$ coincides with $T_{\text{obs}} \leq \sigma^*(z_0)$, while the strict form against the floored count is stricter than against the real exit time off integer τ^* . Two certainty-equivalence laws are audited: an uncorrected law whose index leads the state by one step, $s \mapsto s + 0.1$, and a corrected law with the lead removed; both are read through the index $g(s) = s^2$.

Figures and tables. Every figure and every tabulated entry in this paper is produced by a generation script that first recomputes the quantity from the primitives above, in exact rational arithmetic, and asserts it before drawing or printing; the master verification script of Section 18 chains both scripts and re-derives the tabulated entries independently.

Shared symbols across the companion papers. The companion papers share this vocabulary with paper-local meanings; the letters below carry meanings here that differ there:

Here	In the companions
λ, λ^* : the Farkas and dual certificate weights	the Farkas multiplier and observer decay rate (obstruction calculus); certificate weight vectors (certification; minimax duals)
$\tau^*(z_0)$: the worst-case exit time	observation and revelation times τ (obstruction calculus; certification; minimax duals)
$\sigma^*(z_0)$: the floored exit count	the kernel time variable σ (certification; minimax duals); a noise scale (forecast ladder)

4. A master monotonicity theorem

The audited family is an instance of one abstract object, and its kernel comparisons are instances of one theorem. The finite-system and belief-kernel vocabulary here specializes the **FinSys** and **Wk** definitions of Abaee (2026, An obstruction calculus); the worked-system tuple below supplies its own finite data and audit frame. An *audited tuple* is $\Theta = (X, \mathcal{V}, A, F, \text{info}, \Pi, m)$: a finite state set X with safe set $\mathcal{V}$, action set A , transition F , observation map, admissible policy class Π (per-cell action rules), and register convention m (shared, per-branch, or none); viability is the horizon-stabilized recursion of Section 3.

Proposition 1 (master monotonicity). Fix the audited tuple. Kernel membership is monotone in each of the three enlargements, setwise: (i) *policies*: if $\Pi \subseteq \Pi'$ then $\mathcal{W}_\Pi \subseteq \mathcal{W}_{\Pi'}$ — restriction is antitone (Proposition 4); (ii) *observations*: if info' is finer than info (every fibre of info is a union of fibres of info') then $\mathcal{W}_{\text{info}} \subseteq \mathcal{W}_{\text{info}'}$; (iii) *memory*: a per-branch register dominates the shared register — every shared-register policy is a per-branch policy, so the per-branch kernel contains the shared one. Finer observation maps and larger memories can only enlarge the kernel; all three statements are proved by the same induction on the horizon: each enlargement enlarges the admissible action sets at every cell and horizon level, so a winning continuation transfers upward.

Proof. Induction on h . For $h = 0$ the kernels coincide ($\mathcal{V}$ is fixed). Step: a belief winning at h under the restricted tuple uses, at each cell, an action admissible and winning there; the same choice is admissible under the enlarged tuple (the action set at each cell only grows: more policies choose more actions, finer cells shrink the quantifier over merged branches, a branch-local register relaxes the no-repeat constraint on the other branch), and the continuation is winning at $h - 1$ by the inductive hypothesis. The finite population makes the induction an exact certificate: checks P16–P18 verify the containments setwise on all 36 pairs. $\square$

The audited counts are the theorem’s rows: $24 \leq 26$ and $25 \leq 28$ (observation axis, free protocol), $24 \leq 25$ and $26 \leq 28$ (protocol axis at fixed aggregation), and the per-branch-register kernels (26, 26, 27) dominate the shared ones (24, 26, 25) (memory axis; Proposition 5).

Proposition 2 (stationary policies). Enumerate all deterministic stationary policies — one action per cell, fixed in time: $2^9 = 512$ at full information (nine singleton cells), $2^5 = 32$ at aggregation (sums two to six). At full information the union of the stationary policies’ safe-pair sets is exactly the kernel: memoryless sufficiency holds on the instance, and the 28-pair kernel admits an independent constructive description. At aggregation the union is *empty* — no stationary policy sustains any of the 36 pairs: the merged index is read against states that demand conflicting actions across visits, and every aggregate-viable pair relies on actions scheduled against time. The 26-pair aggregate kernel is therefore entirely a time-varying-policy phenomenon — the register-coupling mechanism in another guise, with the review clock in the register’s place (checks P17–P18; the fixpoint computation of each class’s horizon family independently reproduces all four kernels).

Proof. Both counts are exhaustive enumerations with exact forward simulation under each fixed policy (check P17), cross-validated against the backward recursion by the horizon fixpoint (check P18). $\square$

Strictness is instance-specific: the strict differences are exactly the register-blocked pairs $\{(1, 2), (3, 1)\}$ and $\{(1, 3), (2, 1)\}$ on the protocol axis, the single pair $\{(1, 3), (3, 1)\}$ on the observation axis, with the alternation-forced pair $\{(1, 2), (2, 1)\}$ outside both middle kernels; the incomparability of the middle cells is a shared-register phenomenon, and the decentralized kernel sits outside the chain because unanimity restricts *who* acts, not what the acting policy may do — an orthogonal axis the theorem prices separately (Section 7).

5. The master table

Table 1 is the audit’s master table: all 36 belief pairs against the five kernels — the four policy classes of Section 6 (institutional: one shared compliance register per review; aggregate: one pooled index per branch; full-information codex: pooled registers across branches; full-information: unrestricted state observation) and the decentralized kernel of Section 7. Rows are grouped by the pair’s lexicographically smaller state, written $z_1 z_2$.

The column counts are the audited kernel sizes $|\mathcal{W}_{\text{inst}}| = 24$, $|\mathcal{W}_{\text{agg}}| = 26$, $|\mathcal{W}_{\text{full-codex}}| = 25$, $|\mathcal{W}_{\text{full}}| = 28$, and

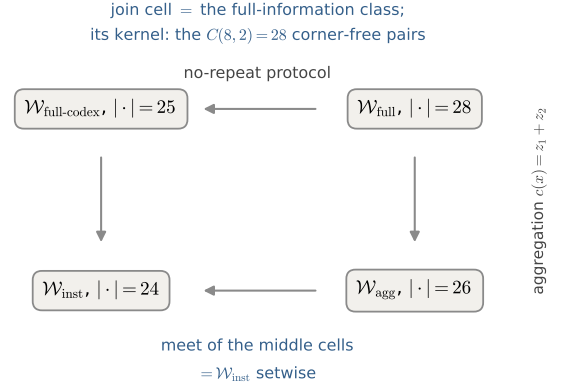


Figure 1: The four policy classes form a two-by-two product of aggregation level and protocol; the audited kernel sizes are 24, 26, 25, 28 of 36. The meet of the middle kernels is the institutional kernel setwise; the class join is the full-information class, whose kernel is the 28 corner-free pairs, one pair beyond the middle-kernel union.

$|\mathcal{W}_{\text{dec}}| = 12$. Figure 1 displays the four policy-class kernels, whose middle cells are incomparable. Two set-level facts are recorded exactly. The meet of the two middle kernels equals the institutional kernel setwise. At class level the join is the full-information class; at kernel level the union of the two middle kernels falls one pair short of the full kernel: exactly the pair $\{(1, 2), (2, 1)\}$ lies in neither middle kernel. The pair is the alternation-forced pair: under full information each branch survives by its own strict alternation — $(1, 2)$ under $1, 2, 1, \dots$ and $(2, 1)$ under $2, 1, 2, \dots$ — which the idealized per-state policy can carry on both branches; aggregation cannot see either requirement (one fibre, no common action), and the no-repeat protocol, whose single previous-action register must serve both branches, forbids the second branch’s required next action once the first branch’s choice occupies the register. The full kernel is exactly the $C(8, 2) = 28$ pairs avoiding the corner $(1, 1)$: the corner’s singleton nonviability accounts for every remaining pair-level loss at full information, as the eight $11|\cdot|$ rows of Table 1 show.

6. Policy classes

The four audited classes combine full information and aggregation with free and no-repeat protocols. Their kernels among the 36 pairs have sizes:

$$|\mathcal{W}_{\text{inst}}| = 24, \quad |\mathcal{W}_{\text{agg}}| = 26, \\ |\mathcal{W}_{\text{full-codex}}| = 25, \quad |\mathcal{W}_{\text{full}}| = 28.$$

The four classes form a two-by-two product of aggregation level and protocol; their kernels have incomparable middle cells: the aggregate kernel contains pairs outside the full-information codex kernel (for instance $\{(1, 2), (3, 1)\}$ and $\{(1, 3), (2, 1)\}$), and the codex kernel contains pairs outside the aggregate (for instance $\{(1, 3), (3, 1)\}$). Within each observation column the free protocol’s kernel contains the no-repeat protocol’s (both inclusions strict); across columns neither middle kernel contains the other. The precise meet-and-join reading is the one recorded in Section 5. Two per-pair readings illustrate the columns of Table 1. The pair $\{(1, 2), (3, 1)\}$ is aggregate-viable but not

Table 1: The master table: membership of all 36 belief pairs in the five audited kernels. I = institutional (aggregation, no-repeat); A = aggregate; C = full-information codex; F = full information; D = decentralized unanimity. A dot marks a non-member; every entry is regenerated by script.

Pair	I	A	C	F	D	Pair	I	A	C	F	D
12 21	.	.	.	F	D	22 31	I	A	C	F	D
12 31	.	A	.	F	.	22 23	I	A	C	F	.
12 23	I	A	C	F	D	22 33	I	A	C	F	D
12 33	I	A	C	F	.	22 32	I	A	C	F	.
12 22	I	A	C	F	.	23 31	I	A	C	F	.
12 32	I	A	C	F	D	23 33	I	A	C	F	.
12 13	I	A	C	F	.	23 32	I	A	C	F	D
13 21	.	A	.	F	.	31 33	I	A	C	F	D
13 31	.	.	C	F	D	31 32	I	A	C	F	.
13 23	I	A	C	F	.	32 33	I	A	C	F	.
13 33	I	A	C	F	D	11 12	.	.	.	.	.
13 22	I	A	C	F	D	11 21	.	.	.	.	.
13 32	I	A	C	F	.	11 31	.	.	.	.	.
21 31	I	A	C	F	.	11 23	.	.	.	.	.
21 23	I	A	C	F	D	11 33	.	.	.	.	.
21 33	I	A	C	F	.	11 22	.	.	.	.	.
21 22	I	A	C	F	.	11 32	.	.	.	.	.
21 32	I	A	C	F	D	11 13	.	.	.	.	.

codex-viable: the free protocol admits it under both observation structures, while the no-repeat protocol excludes it under both — at the second review the thread that records the $(3, 1)$ -branch’s instrument 2 forbids the forced instrument 2 of the successor branch $(2, 1)$, the same shared-register collision that excludes the alternation-forced pair. The pair $\{(1, 3), (3, 1)\}$ is codex-viable but not aggregate-viable: the two branches share the fibre $c = 4$, and no single per-fibre instrument serves both, while full information separates them. Three structural facts behind the table admit short proofs.

Proposition 3 (full information decomposes branchwise). At full information under the free protocol, a two-state belief is viable if and only if each of its singletons is viable; the kernel is exactly the $C(8, 2) = 28$ corner-free pairs.

Proof. Full-information fibres are singletons, so a belief’s step-one action sets are the product of its branches’: a choice is admissible exactly when each branch’s part is, and the recursion decomposes — a belief is viable exactly when both branches are. For the singletons: the corner $(1, 1)$ is nonviable because each instrument drops a coordinate to zero in one step ($u = 1$ forces $z_2 = 0$, $u = 2$ forces $z_1 = 0$, $u = 0$ forces both); every other singleton is viable — the memoryless policy that plays instrument 1 exactly at $z_1 = 1$ and instrument 2 otherwise is safe forever: at $z_1 = 1$ the exclusion of the corner gives $z_2 \geq 2$, so $u = 1$ restores $z_1 = 2$ with $z_2 \geq 1$; at $z_1 \geq 2$, $u = 2$ keeps $z_1 \geq 1$ and regenerates z_2 . $\square$

Proposition 4 (class lattice). Kernel membership is antitone in the policy class: restricting the class can only lose viable beliefs. Hence $\mathcal{W}_{\text{inst}} \subseteq \mathcal{W}_{\text{agg}}, \mathcal{W}_{\text{full-codex}} \subseteq \mathcal{W}_{\text{full}}$; on the audited 36-pair population the two middle inclusions are strict, the meet $\mathcal{W}_{\text{agg}} \cap \mathcal{W}_{\text{full-codex}}$ equals the institutional kernel setwise, and exactly one pair — the alternation-forced pair $\{(1, 2), (2, 1)\}$ — lies in the full kernel outside the middle union.

Proof. Antitonicity — order-reversal between class inclusion and kernel inclusion — is immediate: a policy admissible for a restricted class is admissible for any enlargements, so viability transfers upward. The setwise facts

are finite statements about a 36-element population; they are proved by exhaustive certification — the master table’s 36×5 membership marks are recomputed row-by-row by the paper’s verification script (checks M2 and M5), which is a complete proof for a finite family. $\square$

Proposition 5 (register-semantics invariance). Endow the no-repeat protocol with the pessimistic shared-register semantics: the policy fixes one action per fibre, the register then holds some fibre’s executed action, and the belief survives only if every admissible thread leaves a winning continuation. Because the survival condition quantifies universally over threads, the pessimistic kernels depend on no thread choice: the audited counts 24, 26, 25, 28 are thread-independent by construction. Two further readings are computed for the record. A single-thread rule (one fixed fibre’s action entering the register) is order-sensitive: with the first, or the last, fibre in sorted order the counts are 25, 26, 26, 28 — neither fixed rule yields the audited institutional and codex counts — so the well-defined semantics is the pessimistic one. The optimistic semantics (some thread) is strictly larger (codex 27) and is not the audited reading. If instead each branch carries its own register — one compliance record per unit, threaded forward along the branch — the protocol cost collapses: the institutional kernel equals the aggregate kernel setwise ($26 = 26$), and the codex kernel becomes 27, regaining the alternation-forced pair and losing exactly $\{(1, 3), (3, 1)\}$ relative to full information, with the aggregate kernel contained in it. The incomparability of the middle cells is therefore a shared-register phenomenon; the shared register is the audited institutional object — one compliance register per review across the units managed under the protocol.

Proof. Every clause is a finite statement, computed from the shared primitives of Section 3 and asserted by the verification script: the pessimistic counts (M2b), the optimistic containment (M2c), both single-thread orders, and the per-branch kernels with the $\{(1, 3), (3, 1)\}$ loss (checks P9 and P10). $\square$

Table 2: The 16 law pairs sustaining the audited belief $\{(1, 2), (2, 1)\}$ factor exactly: agency 1’s vote is forced to 1 at $z_1 = 1$ and 2 at $z_1 = 2$, free at $z_1 \in \{0, 3\}$; agency 2’s vote is forced to 2 at $z_2 = 1$ and 1 at $z_2 = 2$, free at $z_2 \in \{0, 3\}$. Every combination sustains — a complete 4×4 product (votes listed as $(l(z=0), l(z=1), l(z=2), l(z=3))$); agency 1’s votes are the row labels, agency 2’s the column labels.

	agency 2			
agency 1	1211	1212	2211	2212
1121	✓	✓	✓	✓
1122	✓	✓	✓	✓
2121	✓	✓	✓	✓
2122	✓	✓	✓	✓

7. Decentralized observation

Two agencies each observe one coordinate and vote; an instrument executes only on agreement. Each agency’s vote is a map from its coordinate’s four values to the two instruments, so the law space has $2^4 \times 2^4 = 256$ law pairs. Unanimity sustains 12 of the 36 pairs (column D of Table 1), and exactly 16 coordinated-viable pairs are lost. The loss instance $\{(1, 2), (3, 1)\}$ is representative: each branch is individually sustainable, and no law pair sustains the belief. For the audited belief $\{(1, 2), (2, 1)\}$, exactly 16 of the 256 law pairs sustain it; the count coincides numerically with the 16 lost pairs, and no identity is implied. The authority restriction is exact:

Proposition 6 (authority restriction). Fixing agency 2’s vote at $z_2 = 1$ to instrument 1 sustains no law pair for the belief $\{(1, 2), (2, 1)\}$: branch (2, 1) exits at its first step under every such law pair. Fixing agency 1’s vote at $z_1 = 1$ to instrument 1 leaves exactly the 16 sustaining law pairs of Table 2, and either whole-law restriction (one agency’s vote fixed to a single instrument at every level) collapses the decentralized kernel to zero.

Proof. On branch (2, 1) at the first step: $z_2 = 1$ must not decline, so agreement on instrument 2 is required; agency 2’s fixed vote at $z_2 = 1$ is 1, so agreement can hold only on instrument 1, which drives z_2 to zero, and disagreement executes $u = 0$, which does the same — the branch exits at step one under every admissible law pair. With agency 1’s vote at $z_1 = 1$ fixed to 1, the sustaining pairs are exactly those with agency 1 forced to 1 at $z_1 = 1$ and 2 at $z_1 = 2$ and agency 2 forced to 2 at $z_2 = 1$ and 1 at $z_2 = 2$ (each forcing is a one-step safety requirement: the low coordinate must regenerate), and the free votes at $\{0, 3\}$ never bind — the 4×4 product of Table 2. A whole-law restriction fixes one agency’s vote everywhere; the other agency’s coordinate then cannot regenerate at the low states where the fixed vote disagrees, and the kernel collapses (verified exhaustively over all 256 law pairs, the whole-law kernels 0 in all four cases; check P12 of the verification script). $\square$

The law space itself is counted before quotienting: a $z = 0$ vote is consultable only on a trajectory that has already left the safe set, so the audited kernel is invariant under the $z=0$ -vote quotient — the effective law space is $2^3 \times 2^3 = 64$ behaviorally distinct law pairs, and the 16 sustaining law pairs of Table 2 comprise 4 distinct behavioral

laws times the 4 never-consulted $z=0$ votes. The sustaining set has an exact product structure (Table 2): on the trajectory alternating between the branches, agency 1 is read only at $z_1 \in \{1, 2\}$ and must vote 1 then 2; agency 2 is read only at $z_2 \in \{2, 1\}$ and must vote 1 then 2; the votes at the unreachable coordinates $z \in \{0, 3\}$ are free, giving $2^2 \times 2^2 = 16$. The restricted kernels are exact. Fixing either agency’s low vote to the regenerating instrument (agency 1 to 1 at $z_1 = 1$, agency 2 to 2 at $z_2 = 1$) leaves the decentralized kernel untouched at 12 — the constraint does not bind. Fixing either agency’s low vote against it (agency 2 to 1 at $z_2 = 1$, agency 1 to 2 at $z_1 = 1$) shrinks the kernel to four beliefs — $\{(2, 3), (3, 2)\}$, $\{(2, 2), (3, 3)\}$, $\{(1, 3), (3, 3)\}$, $\{(1, 3), (2, 2)\}$, up to the symmetry below — and kills the audited target; a whole-law restriction collapses the kernel to 0 in all four cases. The model’s swap symmetry (coordinates and instruments together) interchanges the agencies and maps the audited target to itself, so no agency is privileged: what carries the target is the direction of each forced vote — each branch’s low coordinate must be allowed to regenerate — not the agency holding it. Two modelling choices are declared. The deadlock transition $u = 0$ executes no instrument and both coordinates lose one; and each agency votes on its own coordinate’s current value: the audited protocol is memoryless and decentralized — no agency observes the belief, the other agency’s observation, or anything like common knowledge. The count has a theorem behind it:

Proposition 7 (the decentralized kernel factors statewise). Unanimity executes statewise: the executed instrument at a state depends only on the two votes at that state’s coordinates, and safety is evaluated per state. Hence a belief is sustained if and only if its branches share a common sustaining law pair, $\bigcap_{x \in B} S(x) \neq \emptyset$, where $S(x)$ is the set of law pairs sustaining the singleton $\{x\}$; on the audited population the factored criterion reproduces the joint kernel exactly (the 12 sustained beliefs of the master table), and the 16 sustaining law pairs of the audited belief (Table 2) are invariant under the model’s coordinate-swap symmetry. The obstruction is legible branchwise: a belief is lost exactly when its branches’ sustaining sets do not intersect — as for the representative loss $\{(1, 2), (3, 1)\}$, each branch individually sustainable, no law pair sustaining both.

Proof. The belief trajectory under a law pair is the union of the branch trajectories (statewise execution, statewise safety), so it stays in $\mathcal{V}$ if and only if every branch trajectory does — which is the displayed intersection. Both computations (the factored singleton intersection and the joint 256-pair enumeration) are run on all 36 beliefs and agree setwise (check P20). The symmetry is the swap map of this section’s closing remark applied to the product structure. $\square$

8. Review timing on the hidden-regime grid

The hidden-regime resource model: a single stock held at level $z_0 \geq 1$ above the floor; each review period the unobserved adversarial regime realizes one unit of decline and there is no regeneration (the hold class), so the stock tracks $z(t) = z_0 - t$ and remains safe exactly while $z(t) \geq 1$. The

model is audited on the grid $z_0 \in \{1.0, 1.1, \dots, 4.0\}$ and $T_{\text{obs}} \in \{1, 2, 3\}$, 93 cells. Viability at the deadline holds if and only if $T_{\text{obs}} \leq \tau^*(z_0) = z_0 - 1$, with zero mismatches across the grid. The identity is boundary-inclusive: the viable cells number 33, and the strict form $T_{\text{obs}} < \tau^*(z_0)$ fails exactly on the boundary cells (2, 1), (3, 2), and (4, 3) — those with $z_0 = 1 + T_{\text{obs}}$; against the floored count σ^* the strict form fails on one cell per row $z_0 \geq 2$ — 21 cells, the boundary cells included — which is why the inclusive predicate is the audited identity. Table 3 carries the full grid and Figure 2 displays it. The practical form of the rule: a review must land no later than the worst-case exit time, and a deadline exactly at the worst-case exit is admissible under the audited endpoint convention (closed safe set; safety evaluated at the deadline review); no open-set or alternative-exit-convention claim is made. The audited identity is in fact the restriction of a general law:

Proposition 8 (review timing, general form). On the hold class of the hidden-regime model, for every initial level $z_0 \geq 1$ and every integer deadline $T \geq 0$, the cell (z_0, T) is deadline-viable if and only if $T \leq z_0 - 1$. Accordingly $\sigma^*(z_0) = \lfloor z_0 - 1 \rfloor$ with superlevel sets $\{z_0 \geq 1 + k\}$, $k \in \mathbb{Z}_{\geq 0}$, and the audited 93-cell grid is the restriction to $z_0 \in \{1.0, \dots, 4.0\}$, $T \in \{1, 2, 3\}$.

Proof. Under every admissible hold-class control the adversarial regime realizes the declining branch at each step, so the worst stock declines at unit rate and remains safe exactly through time $z_0 - 1$ — the last safe review, the strict violation occurring immediately after time $z_0 - 1$: deadline-viability forces $T \leq z_0 - 1$. Conversely the hold policy tracks the decline path $z(t) = z_0 - t$ branchwise, which stays admissible exactly on $[0, z_0 - 1]$ and terminates at the floor, which is safe because the safe set is closed; hence $T \leq z_0 - 1$ is viable, boundary included. The set identity $\{T : T \leq z_0 - 1\} = \{T : T \leq \sigma^*(z_0)\}$ (integer T , all z_0) — equivalently $\{T < z_0 - 1\} = \{T \leq \sigma^*(z_0)\}$ off the integers, with the boundary failure at integer $z_0 - 1$ — gives the displayed supremum. $\square$

Proposition 9 (hitting times on regime graphs). Let the adversary choose a regime path $r_1 r_2 \dots$ through a finite regime graph with per-step damages $d_r \geq 0$, subject to the graph’s adjacency constraint. The worst cumulative damage by time t is the max-plus recursion $D_t = \max_{\text{paths}} \sum_{s \leq t} d_{r_s}$ — the standard timed-event-system algebra (Baccelli et al., 1992) — and the cell (z_0, T) is deadline-viable if and only if $D_T \leq z_0 - 1$ (the audited endpoint convention). Two exact instances: with free switching the adversary plays the worst regime every step, $D_t = t \cdot \max_r d_r$, recovering Proposition 8 on the audited grid; with two regimes of damages 1 and $1/2$ forced to alternate, the worst damage is $D_{2k} = 3k/2$ and $D_{2k+1} = 3k/2 + 1$, and the re-audited 93-cell grid has exactly 43 viable cells (21, 16, 6 at $T = 1, 2, 3$), zero mismatches between the closed form and path enumeration. The max-plus dynamic program equals path enumeration exactly on a three-regime graph with a forbidden self-loop (horizons one through five in the deposited P21 check).

Proof. Deadline-viability is safety of the worst adversarial path to the deadline, which is the max-plus evaluation of the damage sum; free switching decouples the steps, giving

Table 3: The timing grid, printed in full (93 cells): viability at the deadline against $\sigma^*(z_0) = \lfloor \tau^*(z_0) \rfloor$ (the deadline count; the identity columns use the inclusive predicate, on which the floored and real-valued forms coincide). The checkmark of column T_{obs} means the cell is viable; the identity column “id.” is the predicate $T_{\text{obs}} \leq \sigma^*(z_0)$ and column “cell” the audited viability; the two agree on all 93 cells (zero mismatches; 33 viable cells). The circled cell of Figure 2 is (2.0, 1): viable, on the boundary; the strict form disagrees exactly on the boundary cells (2, 1), (3, 2), (4, 3).

z_0	σ^*	$T_{\text{obs}} = 1$		$T_{\text{obs}} = 2$		$T_{\text{obs}} = 3$	
		id.	cell	id.	cell	id.	cell
1.0	0	.	.	.	.	.	.
1.1	0	.	.	.	.	.	.
1.2	0	.	.	.	.	.	.
1.3	0	.	.	.	.	.	.
1.4	0	.	.	.	.	.	.
1.5	0	.	.	.	.	.	.
1.6	0	.	.	.	.	.	.
1.7	0	.	.	.	.	.	.
1.8	0	.	.	.	.	.	.
1.9	0	.	.	.	.	.	.
2.0	1	✓	✓	.	.	.	.
2.1	1	✓	✓	.	.	.	.
2.2	1	✓	✓	.	.	.	.
2.3	1	✓	✓	.	.	.	.
2.4	1	✓	✓	.	.	.	.
2.5	1	✓	✓	.	.	.	.
2.6	1	✓	.	.	.	.	.
2.7	1	✓	.	.	.	.	.
2.8	1	✓	.	.	.	.	.
2.9	1	✓	.	.	.	.	.
3.0	2	✓	✓	.	.	.	.
3.1	2	✓	✓	.	.	.	.
3.2	2	✓	✓	.	.	.	.
3.3	2	✓	✓	.	.	.	.
3.4	2	✓	✓	.	.	.	.
3.5	2	✓	✓	.	.	.	.
3.6	2	✓	✓	.	.	.	.
3.7	2	✓	✓	.	.	.	.
3.8	2	✓	✓	.	.	.	.
3.9	2	✓	✓	.	.	.	.
4.0	3	✓	✓	✓	✓	.	.

$t \cdot \max_r d_r = t$ on the audited damages and hence the audited identity. Under alternation the two interleavings give $D_{2k} = k(1 + \frac{1}{2}) = 3k/2$ and $D_{2k+1} = (k+1) + \frac{k}{2}$; the grid recomputation enumerates both interleavings per cell (check P21). The dynamic-programming (DP) evaluation over (last regime, step) is exact rational arithmetic and is asserted equal to explicit path enumeration. $\square$

9. Monitoring adequacy

Let $R(x)$ denote the safe actions of state x : those keeping x in $\mathcal{V}$ for one step. On the audited target, $R((1, 2)) = \{1\}$, $R((2, 1)) = \{2\}$, $R((2, 2)) = \{1, 2\}$, and $R((3, 1)) = \{2\}$. A partition of a target belief is *adequate* when every fibre has a nonempty common safe-action set. On the four-state target $\{(1, 2), (2, 1), (2, 2), (3, 1)\}$, 7 of the 15 partitions are adequate, and two minimal adequate partitions with two cells exist. Table 4 carries the complete census with each fibre’s common safe-action set; Figure 3 displays it.

The maximal subsets with a nonempty common safe-action set are $\{(1, 2), (2, 2)\}$ and $\{(2, 1), (2, 2), (3, 1)\}$. They overlap in (2, 2), so no partition into maximal common-action sets exists: coarsening to maximal fibres

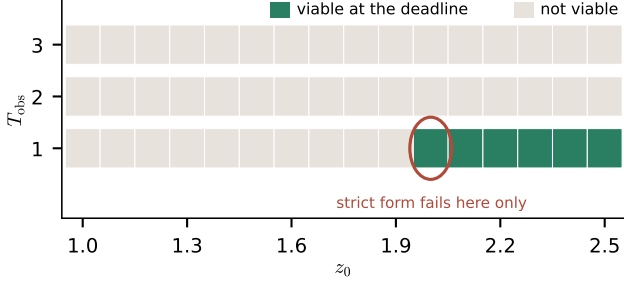


Figure 2: The timing grid. Cells viable at the deadline (green) satisfy $T_{\text{obs}} \leq \tau^*(z_0)$; the identity is boundary-inclusive, and the strict form $T_{\text{obs}} < \tau^*(z_0)$ fails on three boundary cells, (2.0, 1), (3.0, 2), (4.0, 3) (the first is circled); against the floored count $\sigma^*(z_0) = \lfloor z_0 - 1 \rfloor$ the strict form fails on 21 of the 33 viable cells, which is why the inclusive predicate is the audited identity.

Table 4: The fifteen-partition census on the target $\{(1, 2), (2, 1), (2, 2), (3, 1)\}$ (states written $z_1 z_2$; fibres separated by “;”). A partition is adequate when every fibre’s common safe-action set (brackets; $\cdot$ = empty) is nonempty. Seven partitions are adequate; the minimal adequate two-cell partitions are rows 6 and 7.

Partition	Adequate	Common safe actions
12 21 22 31	no	$[\cdot]$
12 21 31 ; 22	no	$[\cdot \ 12]$
12 22 31 ; 21	no	$[\cdot \ 2]$
12 31 ; 21 22	no	$[\cdot \ 2]$
12 31 ; 22 ; 21	no	$[\cdot \ 12 \ 2]$
21 22 31 ; 12	yes	$[2 \ 1]$
21 31 ; 12 22	yes	$[2 \ 1]$
21 31 ; 22 ; 12	yes	$[2 \ 12 \ 1]$
22 31 ; 12 21	no	$[2 \cdot]$
22 31 ; 21 ; 12	yes	$[2 \ 2 \ 1]$
31 ; 12 21 22	no	$[2 \cdot]$
31 ; 12 22 ; 21	yes	$[2 \ 1 \ 2]$
31 ; 21 22 ; 12	yes	$[2 \ 2 \ 1]$
31 ; 22 ; 12 21	no	$[2 \ 12 \cdot]$
31 ; 22 ; 21 ; 12	yes	$[2 \ 12 \ 2 \ 1]$

is not an available design, and the adequate partitions found by the census are the design options. The criterion is one-step monitoring adequacy; dynamic viability is not claimed from it (the kernels are computed by the full recursion). The census excludes instrument 0 without loss: $F_0 \leq F_1$ and $F_0 \leq F_2$ pointwise with the safe set an up-set (verified in the frame section) give every one-step-safe fibre a safe instrument in $\{1, 2\}$. Two consequences follow. Pairwise intersection of safe-action sets is necessary for fibre adequacy; in the two-instrument audited universe it is also sufficient (Helly number two: a pairwise-intersecting family of subsets of a two-point action set has a common action), so every inadequate fibre in the census fails with a literal empty intersection, while in action sets of three or more elements pairwise intersection ceases to suffice ($\{1, 2\}, \{2, 3\}, \{1, 3\}$); and an observation map merging states with disjoint safe-action sets obstructs at that information state while every branch remains individually viable — the remedy is a fibre split, not a finer threshold on the same index. The census makes both visible at the level of individual fibres: in each inadequate partition of Table 4 the failing fibre is exactly one merging (1, 2) — whose safe-action set is the singleton $\{1\}$ —

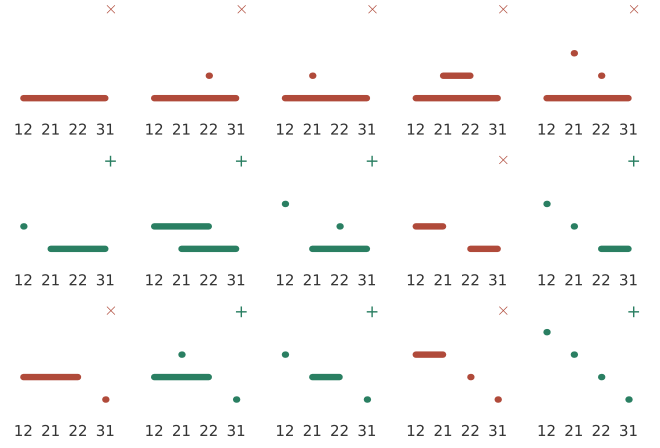


Figure 3: The census. Each panel draws one partition: a bar over a group of states is a fibre (green when its common safe-action set is nonempty, red when empty), and the margin marks adequacy (+/×). The two adequate two-cell partitions are panels 6 and 7 of the middle row.

with a state whose safe actions exclude 1, i.e. (2, 1) or (3, 1), and each adequate partition merges (1, 2) with neither. The census is itself an instance of a structure:

Proposition 10 (adequacy as an incompatibility-graph criterion). Let G be the incompatibility graph of the target: vertices are states, with an edge (x, x') when $R(x) \cap R(x') = \emptyset$. A partition is adequate if and only if every fibre is an independent set of G ; on the audited target the edges are exactly $\{(1, 2)-(2, 1), (1, 2)-(3, 1)\}$, so the adequate partitions are the partitions into independent sets, counted by inclusion–exclusion as $15 - (5 + 5 - 2) = 7$ (the two edge events overlap in exactly the 2 partitions whose fibre contains (1, 2), (2, 1), (3, 1) jointly), and the adequate two-cell partitions are exactly the graph’s two proper 2-colorings — rows 6 and 7. In an a -action universe, a fibre fails adequacy only through a subfamily of at most a states with empty common intersection: the Helly number of the family of subsets of an a -element action set is a (verified exhaustively for $a = 2$ and $a = 3$, where the minimal failing family $\{\{1, 2\}, \{2, 3\}, \{1, 3\}\}$ attains it); pairwise intersection suffices only for $a = 2$, and for convex m -dimensional action domains the bound is $m + 1$ (Proposition 13).

Proof. A fibre is adequate exactly when its states’ safe-action sets intersect, which is the complement of an edge exactly on two-state fibres and, since the audited universe has Helly number two, for all fibres (Proposition 13 is the m -dimensional analogue). The edge count is the two disjointness facts above. The inclusion–exclusion count: partitions whose fibres include the edge (1, 2)–(2, 1) are partitions of the three objects $\{(1, 2)(2, 1)\}, (2, 2), (3, 1)$ — five; symmetrically five for the other edge; both edges force the fibre $\{(1, 2), (2, 1), (3, 1)\}$ — two partitions (with and without (2, 2) merged). A two-cell adequate partition is a bipartition into independent sets, i.e. a proper 2-coloring of the star; there are exactly two. The Helly statements are finite computations over all subfamilies of the 2^a subsets of an a -element set (exact for $a = 2, 3$; check P19). $\square$

10. Regime uncertainty

Two regimes alternate: an adversary chooses at the end of step $t - 1$ the regime realized at step t . The model, explicit: the controller acts one instrument $u \in \{1, 2\}$ per step (coordinate u regenerates by one, cap 3), and the acting regime deals one unit of damage to its own coordinate (floor 0); there is no other change. With the regime frozen and announced, the controller drives the state to a fixed point — acting the damaged coordinate holds it wherever its level is below the cap, and one step of the other instrument frees a capped coordinate — so the corner $(1, 1)$ is viable under each regime separately, and each frozen kernel comprises all 36 belief pairs. Mode-blind operation, with the regime read only through the state, yields a kernel of exactly 28 — every pair containing the corner is lost (from $(1, 1)$ either instrument regenerates one coordinate while the adversarial damage zeroes the other), no merged-pair loss exists, and the loss is generated entirely at the singleton level. The mode-blind kernel agrees setwise with the full-information kernel of Section 6; the agreement is coincidental, arising under different damage routings, and no identity is claimed. All three counts (frozen 36 per regime, corner viable under each, mode-blind 28) are machine-verified from these primitives (check P13 of the verification script).

11. The continuous benchmark

A state is a pair of stocks $(x_1, x_2) \geq 0$ with a demand floor on each; the aggregate is $Y = x_1 + x_2$, and the Y -fibre is the set of states with that aggregate. One control set $U = \{u \geq 0 : u_1 + u_2 \geq 2\}$ faces the floors through Y -dependent caps $\text{cap}_1(Y) = 3/2 - (Y - 2)/10$ and $\text{cap}_2(Y) = 59/50 - (Y - 2)/10$; an allocation $u \in U$ is feasible on the fibre when $u_i \leq \text{cap}_i(Y)$ — fibre feasibility, not transition dynamics, is this section’s audited object. The critical aggregate is $Y^* = 27/5$: the cap sum equals 2 exactly there. Table 5 evaluates the caps on the audited aggregates and Figure 4 displays them. The $Y = 5$ fibre is viable with witness $(6/5, 4/5)$; the $Y = 6$ fibre is obstructed with cap sum $47/25 < 2$, certified by the dual measure $(1/2, 1/2)$ on the two active-floor atoms with margin $3/50$. The crossover is again a general law:

Proposition 11 (exact benchmark crossover). For every aggregate $Y \geq 2$ with $\text{cap}_i(Y) \geq 0$, the Y -fibre admits a feasible input (some $u \in U$ with $u_i \leq \text{cap}_i(Y)$) if and only if $\text{cap}_1(Y) + \text{cap}_2(Y) \geq 2$, which holds if and only if $Y \leq 27/5$. For $Y > 27/5$ the ℓ_1 -normalized dual measure $(1/2, 1/2)$ on the two cap rows certifies the obstruction with the exact margin $\frac{1}{2}(2 - \text{cap}_1(Y) - \text{cap}_2(Y)) = (Y - 27/5)/10$. At $Y = 27/5$ the margin is zero and the fibre remains feasible at the boundary.

Proof. If the cap sum is at least 2, the input $u = (\text{cap}_1(Y), 2 - \text{cap}_1(Y))$ is nonnegative ($\text{cap}_1(Y) \leq 3/2 < 2$ on $Y \geq 2$), lies in U (its coordinate sum is 2), and respects the caps ($2 - \text{cap}_1(Y) \leq \text{cap}_2(Y)$ is the cap-sum inequality). Conversely any $u \in U$ under the caps has $2 \leq u_1 + u_2 \leq \text{cap}_1(Y) + \text{cap}_2(Y)$, so a cap sum below 2 is infeasible. The cap sum is the affine function $67/25 - (Y - 2)/5$, equal to 2 exactly at $Y = 27/5$ and declining beyond; the dual measure gives $\lambda^\top(\text{cap} - u) =$

Table 5: The benchmark’s cap arithmetic on the audited aggregates (exact rationals). The cap sum crosses the demand level 2 exactly at $Y^* = 27/5$; at $Y = 6$ the cap sum falls $3/25$ short of 2, and the ℓ_1 -normalized dual measure $(1/2, 1/2)$ certifies the obstruction with margin $3/50$.

Y	cap_1	cap_2	sum	reading
4	13/10	49/50	57/25	above
9/2	5/4	93/100	109/50	above
5	6/5	22/25	52/25	viable (witness $(6/5, 4/5)$)
27/5	29/25	21/25	2	critical
11/2	23/20	83/100	99/50	below
6	11/10	39/50	47/25	obstructed, margin $3/50$

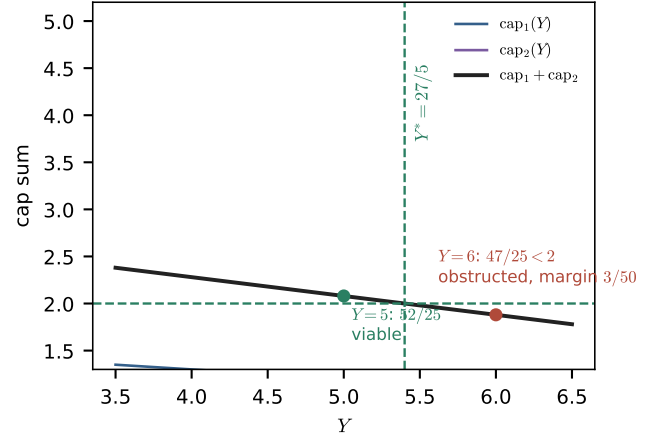


Figure 4: The benchmark. Both caps decrease linearly in the aggregate Y ; their sum crosses the demand level 2 exactly at $Y^* = 27/5$. The $Y = 5$ fibre is viable (green), the $Y = 6$ fibre is obstructed with margin $3/50$ (red).

$(\text{cap sum})/2 - (u_1 + u_2)/2 \leq (\text{cap sum})/2 - 1$, the stated margin. $\square$

The derivation of the caps from a two-stock resource model, and the fibre-width formula $\Delta x_1(Y) = Y - 4$, are carried in the dual-certificate companion (Abae, 2026, Minimax dual certificates); this section owns the per-aggregate audit and the radius reading.

12. Static duality

Static duality for the benchmark’s two floor rows, abstracted to drift rows on $[0, 1]$: the max–min equals the min– λ max at value $-1/10$, with multiplier $\lambda = (1/2, 1/2)$ the ℓ_1 -normalized Farkas pair of the calculus’s worked certificate (Abae, 2026, An obstruction calculus). Figure 5 displays the instance: under the ℓ_1 -normalized multiplier the expected drift is constant $-1/10$ on all of $[0, 1]$. Convexity is load-bearing: with $U = \{0, 1\}$ and rows $(-1, +1)$, $(+1, -1)$, the common set is empty, the max–min is -1 , and the min-multiplier maximum is 0. Under convexity the witness is sparse:

Proposition 12 (static minimax duality on the two-floor instance). On the obstructed instance with drift rows $\psi_1(u) = 2/5 - u$, $\psi_2(u) = u - 3/5$ on $U = [0, 1]$,

$$\max_{u \in [0, 1]} \min_i \psi_i(u) = \min_{\lambda \in \Delta_2} \max_{u \in [0, 1]} (\lambda_1 \psi_1 + \lambda_2 \psi_2) = -\frac{1}{10},$$

with the unique saddle point $(u^*, \lambda^*) = (1/2, (1/2, 1/2))$;

under λ^* the expected drift is constant $-1/10$ on all of $[0, 1]$.

Proof. The min envelope $\min_i \psi_i(u)$ is concave and piecewise linear with its peak at the crossing $2/5 - u = u - 3/5$, i.e. $u^* = 1/2$, value $-1/10$. For $\lambda \in \Delta_2$ the pooled drift $(2\lambda_1 - 1) \cdot (-u) + \text{const}$ is affine with slope $\lambda_2 - \lambda_1 = 1 - 2\lambda_1$; it is constant exactly at $\lambda^* = (1/2, 1/2)$, where the value is $\frac{1}{2}(2/5) + \frac{1}{2}(-3/5) = -1/10$ for every u . At any other λ the maximum over $[0, 1]$ is attained at an endpoint; writing $A(\lambda)$ for the $u = 0$ endpoint value, the two endpoint values are $A(\lambda)$ and $A(1 - \lambda)$, and the averaging bound $\max(A(\lambda), A(1 - \lambda)) \geq \frac{1}{2}(A(\lambda) + A(1 - \lambda)) = -\frac{1}{10}$ — the average is exactly the constant value, for every λ — with equality only at λ^* : the convexity of the max, not symmetry alone, locates the minimum. Weak duality ($\max \leq \min$ -max) holds unconditionally; the exhibited pair attains both sides at $-1/10$, which is therefore the exact common value and the unique saddle. $\square$

Proposition 13 (sparse obstruction witnesses). An infeasible finite family of affine constraints on a convex domain in $\mathbb{R}^m$ has an infeasible subfamily of at most $m + 1$ members, and $m + 1$ is tight. The instance realizes the bound: the three constraints $\{u_1 \geq 1, u_2 \geq 1, u_1 + u_2 \leq 1\}$ are pairwise feasible, jointly infeasible, and no two of them certify the obstruction.

Proof. Intersect each constraint’s halfspace with the domain: an infeasible finite family of convex sets in $\mathbb{R}^m$ has an infeasible subfamily of size at most $m + 1$ (Helly’s theorem), and $m + 1$ is tight: the reversed halfspaces $\{u_i \geq 1\}$ and $\sum_i u_i \leq m - 1$ in $\mathbb{R}^m$ meet m at a time and never all $m + 1$: with the sum row omitted take every $u_i = 1$; with the row $u_j \geq 1$ omitted take $u_j = 0$ and the remaining $u_i = 1$. The full family demands $m \leq m - 1$, a contradiction for every $m \geq 1$. For the instance, each pair is feasible — at $(1, 1)$, $(1, 0)$, and $(0, 1)$ respectively — while all three give the contradiction $2 \leq 1$; the verification script verifies each pair at its printed feasible point and the triple by the exact contradiction. $\square$

The Helly-tight family of three constraints $\{u_1 \geq 1, u_2 \geq 1, u_1 + u_2 \leq 1\}$ — no two of them already infeasible — attains the obstruction with $m + 1 = 3$ constraints. The feasible contrast at rows $(2/5 - u, u - 1/5)$ carries value $+1/10$ on both sides of the saddle: a certificate exists exactly where the value is negative, and at a zero value the instance is boundary-critical with no strictly positive margin available. Weak duality ($\max - \min \leq \min - \max$) holds unconditionally. Strong duality holds where its hypotheses are verified: the control set compact *and convex*, the rows affine (or concave) in u , and the label family finite — as in the static instance above and the finite-horizon static-observation classes of the calculus (Abae, 2026, An obstruction calculus). Both hypotheses are load-bearing: the paper’s own nonconvex example ($U = \{0, 1\}$, rows $(-1, +1)$, $(+1, -1)$) is compact, finite, and row-affine, yet gaps $(-1$ against $0)$; and continuity of the rows does not repair convexity, as the nonconvex rows example above shows — concavity of the rows in u , not continuity, is what the minimax argument consumes.

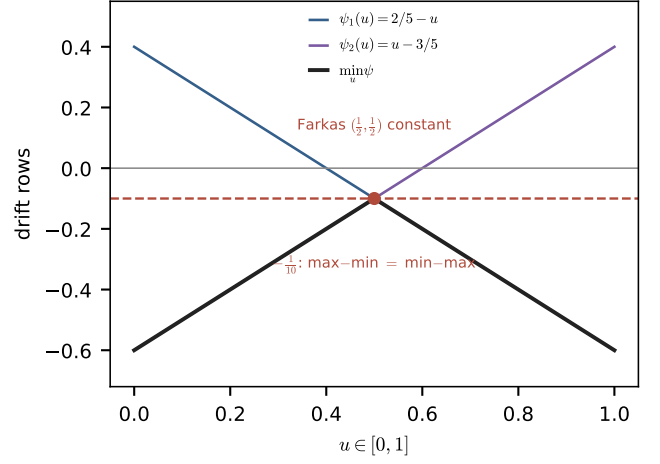


Figure 5: Static duality on the two-floor instance. The two drift rows cross at $u = 1/2$; the max-min equals the min-multiplier maximum at $-1/10$, and the ℓ_1 -normalized Farkas pair $(1/2, 1/2)$ makes the expected drift constant at the value.

Table 6: The certainty-equivalence drift audit: $g(s + 1/10) - g(s) = s/5 + 1/100$ exactly; the corrected law’s drift is 0 at every probed state.

s	uncorrected	corrected	increment per 1/4
1	21/100	0	
5/4	26/100	0	1/20
3/2	31/100	0	1/20
7/4	36/100	0	1/20
2	41/100	0	1/20

13. The certainty-equivalence drift audit

For the index $g(s) = s^2$ the uncorrected law’s one-step drift is $g(s + 1/10) - g(s) = s/5 + 1/100$ on $[1, 2]$: increasing, with minimum 21/100 at $s = 1$ and maximum 41/100 at $s = 2$ (Table 6); the corrected law’s drift is identically zero — by construction, the corrected law being the lead removed. A biased observation map misreads the state by a fixed one-step lead; on any decision model conditioned on the map, that misreading is the mechanism through which a nonempty kernel could empty, and the drift audit bounds it exactly — the remedy is structural correction of the map, not a better sensor.

14. Multiple floors

With several binding floors the Farkas certificate obstructs jointly: a single certificate over all floors certifies nonviability where per-floor certificates do not compose. On the audited benchmark’s two floors (Section 11), the minimal infeasible subsystem carries explicit witnesses — the Helly-tight triple of Section 12 — and the corner $(1, 1)$ is dynamically nonviable as in the shared system. *Quantifier and dual form.* Joint maintenance of several floors is the quantifier $\exists u \forall j \in I(x)$ — one control satisfying every active floor — not $\forall j \exists u_j$; per-floor certificates do not compose. The gap is visible in dimension one: on $U = [-1, 1]$ the floors $g_1(u) = u - \frac{1}{2} \geq 0$ and $g_2(u) = -u - \frac{1}{2} \geq 0$ are each separately maintainable, yet no common control maintains both, and the ℓ_1 -normalized multiplier $\lambda = (\frac{1}{2}, \frac{1}{2})$ certifies the joint infeasibility with $\sup_{u \in U} (\lambda_1 g_1 + \lambda_2 g_2)(u) = -\frac{1}{2} < 0$ — a strict-separation

dual certificate of exactly the Proposition 13 shape, with $m + 1 = 2$ constraints necessary in dimension $m = 1$ (each single floor is feasible). Scalar aggregation inherits the gap: a positive weighted average $\sum_j \lambda_j h_j$ can conceal a violated component, and a multiplier inequality $\sum_j \lambda_j D h_j \geq 0$ does not imply that every active drift is nonnegative; the ℓ_1 -normalized dual measure of Section 11 certifies the aggregate shortfall exactly because both cap rows are charged jointly. In the audited convex static examples, infeasibility is certified by a multiplier supported on at most $m + 1$ active constraints; no such bound is claimed for nonconvex controls or dynamic-policy images. Two further objects are recorded for subsequent study: a first-breach timing bound for multi-floor systems, and a corner-cone reading of the infeasibility region.

15. Robustness margins

Each certificate carries a margin, and each margin is a robustness radius — the size of the perturbation the certificate’s verdict tolerates.

Proposition 14 (margins as radii). (i) *Benchmark*: at $Y = 6$ the obstruction survives any per-cap increase $\delta < 3/50$ of both caps and only there: the certificate’s margin $3/50$ is exactly the per-cap robustness radius (feasibility returns at $\delta = 3/50$). (ii) *Timing*: the boundary cells have zero slack ($z_0 - 1 - T_{\text{obs}} = 0$): any adversarial damage above unit rate flips them — the inclusive convention is exactly the statement that the audited identity holds on zero-margin cells. (iii) *Drift*: the corrected law’s zero drift has the sign of the correction error ε ($2\varepsilon + \varepsilon^2 = \varepsilon(2 + \varepsilon)$ at $s = 1$): the sign-certificate is robust for every $\varepsilon \in (-2, 0) \cup (0, \infty)$. (iv) *Decentralization*: each of the four forced votes of the audited belief is knife-edge — flipping that single vote sustains nothing (zero margin, exact); the protocol’s value is carried by four coordinates, each of which is load-bearing.

Proof. (i) feasibility at $Y = 6$ under uniform increases δ is $\frac{47}{25} + 2\delta \geq 2 \iff \delta \geq \frac{3}{50}$ — exact arithmetic, checked at $\delta = \frac{3}{200}$ (infeasible) and $\delta = \frac{3}{50}$ (feasible). (ii) at $(z_0, T) = (2, 1)$ the worst damage under rate $1 + \varepsilon$ is $1 + \varepsilon > 1 = z_0 - 1$. (iii) the sign evaluation is the displayed factorization. (iv) for each forced cell, the count of sustaining law pairs with that vote flipped is zero (exhaustive over the 256 pairs; check P22). $\square$

16. Design rules

The rules hold on the audited systems — each an exact census identity or a proved general law, tagged by its evidence — and transfer as default guidance only at that evidence level. *Timing*: (proved general law, Proposition 8) reviews are boundary-inclusive — $T_{\text{obs}} \leq \tau^*(z_0)$, the deadline at worst-case exit admissible under the audited endpoint convention (Section 8). *Coarseness*: (exact census, Section 9) a fibre merging states with disjoint safe-action sets obstructs at that information state; the remedy is a fibre split, not a finer threshold on the same index (Section 9). *Aggregation*: (exact census, Sections 6 and 9) the aggregate index’s fibres cross the safe boundary — some mix states of conflicting state-safety verdict — and the aggregate kernel (26 against the full 28) certifies only

a certainly-safe inner approximation; the certainly-safe set is the sound report. *Bias*: (exact audit, Section 13) the uncorrected law’s drift is at least $21/100$ on a 101-point grid of $[1, 2]$, while the corrected law’s drift is identically zero; a fixed bias is exactly correctable, and the remedy is structural correction of the map, not a better sensor. The decentralized audit adds a final rule: (proved — the restricted kernel is zero, Proposition 6) a whole-law authority restriction on one agent’s vote collapses the sustainable set outright, so protocol design precedes instrument tuning (Section 7).

17. Mechanized verification

The master monotonicity statement of Proposition 1 is formalized in the project’s Lean 4 layer, in namespace `Formalizations.WS` (`WS_WorkedSystems`). Its statement imports the earlier `Formalizations.P1_Obstruction` definitions `FinSys` and `Wk` (Abaee, 2026, An obstruction calculus); this is a formal vocabulary dependency, not identity of the papers’ results. In that vocabulary, `master_monotone` and `monotone_action_axis` prove monotonicity under pointwise inclusion of the admissible-action predicate in an abstract finite system. The separate observation-refinement declaration `Wk_mono_obs` requires its own refinement hypotheses. Neither signature encodes the audited no-repeat register threads or proves the table counts.

No admitted gaps. A comment-stripped source scan of the whole project finds no `sorry`, `admit`, `axiom` or `constant` declaration anywhere. This scans explicit source syntax, not generated axiom footprints; the pinned build succeeds and the project’s P3 support/value examples retain three `native_decide` uses (nine generated axiom names).

Scope, stated plainly. What is machine-checked is an action-inclusion monotonicity statement and, separately, a finite observation-refinement statement under their exact hypotheses; the register/count cases are not Lean instances. **The audits themselves are not formalized:** the counts of survivable belief pairs, the kernel sizes under policy-class restrictions, the partition-adequacy census and the dual certificates are computations performed by the deposited scripts in integer and rational arithmetic, and are regenerated rather than proved. This is the mirror image of the usual arrangement, and it is worth being exact about it: the action-inclusion and observation-refinement implications are proved under their respective hypotheses, while the register encoding and numeric entries are computed separately.

18. Verification methods

Every count, witness, identity, bound, figure, and tabulated entry in this paper is checked by deterministic arithmetic scripts using exact integer and rational arithmetic and the standard library; the separate PDF figure renderer uses `matplotlib`. The master verification script recomputes the counts of Sections 5–10 from the shared system’s primitives — the transition, the fibre partition function, and the backward recursion with the no-repeat protocol threaded as a parameter — re-derives every tabulated entry of Tables 1, 2, 3, 4, 5, and 6 independently, enumerates

the census of Section 9, evaluates the benchmark identities of Section 11 in rational arithmetic, asserts the drift bounds, and checks that each headline number appears in the text. The figure-and-table generation script recomputes and asserts each plotted or printed quantity from the same primitives before drawing; the two scripts agree on every shared quantity. The audited identities are additionally covered by the per-system verification scripts of the companion papers (Abaee, 2026, An obstruction calculus; see the Code availability statement). Checks P17–P18 expose a distinction between stationary policies and the central recursions: enumerating all $2^5 = 32$ aggregate stationary policies gives an empty union of safe-pair sets, while the aggregate time-varying kernel contains 26 pairs; enumerating all $2^9 = 512$ full-information stationary policies reproduces its 28-pair kernel. The separate horizon fixpoint recomputes the audited kernels setwise. Memoryless sufficiency here is a full-information result, not an aggregate one. The recursion’s cost is measured exactly: the memoized backward recursion over the 36-pair population uses the distinct-subproblem counts printed by the script (states $\times$ horizon $\times$ register per class), the policy enumerations use 32 and 512 candidates, the decentralized census 256 law pairs (64 behavioral), the adequacy census 15 partitions, and the two timing grids 93 cells each.

19. Conclusion

Exact audits bind the obstruction calculus to its worked systems: each mechanism acquires a count, a witness, and a boundary. The timing certificate fires everywhere beyond $z_0 - 1$ and nowhere at the boundary; the adequate monitoring designs on the audited target are enumerable, and no coarsest one exists; policy-class value drops unequally across aggregation and protocol restrictions, with the two middle cells incomparable, the meet of their kernels institutional setwise, and exactly one pair — the alternation-forced pair — beyond both; unanimity costs a fixed and identified set of beliefs, sustained by exactly the 4×4 product of free votes, recoverable only while both votes remain free; regime blindness loses exactly the corner’s pairs, while no frozen regime loses any; and the continuous benchmark separates the viable from the obstructed at $Y^* = 27/5$ with rational certificates on both sides. The incomparability of the middle cells is a shared-register phenomenon: with per-branch registers the protocol cost collapses (Proposition 5), which is precisely why the shared register is the audited institutional object. The design rules — boundary-inclusive reviews, fibre splits over threshold tuning, certainly-safe reporting under aggregation, structural repair of bias, and protocol design before instrument tuning — are the audit’s operational content, and the master table is its evidence: every count in the paper is one column of one table, every entry of which is regenerated by script. These examples do not constitute a general taxonomy of viability failures; they supply exact counterexamples, finite censuses, and reproducible benchmarks that separate mechanisms often conflated in qualitative discussion. Read through the master theorem, the audit is a small theory: the five mechanisms — common-action failure, memory/register coupling, finite-time boundary exit, decentralized information loss, and joint convex infeasibility — each carry

a certificate, a monotonicity status under the theorem’s three axes, a requirement-level characterization (statewise factoring for unanimity, incompatibility-graph adequacy for fibres, max-plus hitting times for regimes), and a robustness radius (3/50 per cap on the benchmark, zero on the boundary cells and the forced votes, sign-robustness on the drift). Refinement completeness on the continuous side is the matching axis: the certification companion’s meshes, stored scenarios, and belief cells tighten its sandwich exactly as the three axes here enlarge the kernels, with positivity non-monotone on both sides (Abaee, 2026, Computational viability certification)

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Competing interests

None.

Data availability

No external data were used.

Code availability

The verification script `paper2_worked_systems_v17_verification.py` and the figure-and-table generation script `paper2_worked_systems_figures.py` are publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); they reproduce all counts, witnesses, bounds, figures, and tabulated entries verbatim.

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